Unit 5

Hongmei pang

- Style and function of Results and Discussion
- Tables+figures
- Building coherence
- Collocation

Things should always be made as simple as possible and no simpler

– Albert Einstein, 1879–1955

Results

- Not all journals require a separate Results section.
- It can be integrated with the Discussion, under the section title **Results and Discussion**.

Results

Function

- To **objectively present** your key results, **without interpretation**, in an **orderly and logical sequence** using both **text and illustrative materials** (Tables and Figures).

Results

Template

- Experimental purpose
- Approach
- Results

Each subheading may contain a combination of :

- texts: to explain about the research data
- figures: to display the research data and to show trends or relationships
- tables: to represent a large data and exact value

Decide on the best way to present your data — in the form of text, figures or tables (Hofmann, 2013).

Results

Style

- Write the text of the Results section **concisely and objectively**.
- Use the **past tense**.
- Organize the results in a sequence that highlights the answers to your research questions.

Results

Style

1.**We found that doctors viewed the NHS as** having failed to provide adequate services.

personal

- less formal
- helps the reader to become more involved
- more enjoyable to follow and easier to digest

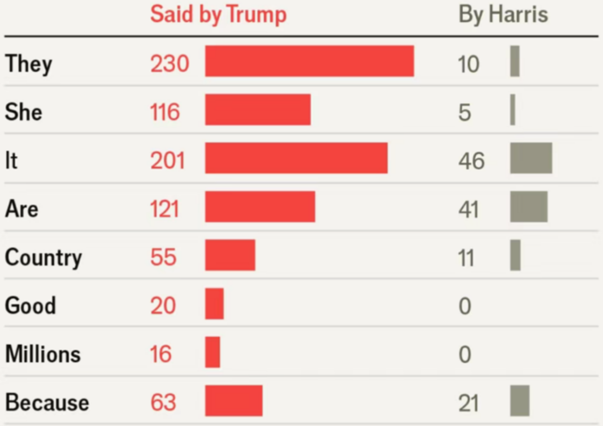
2. **There was a perceived** failure of the NHS to provide adequate services.

impersonal

- adds an element of objectivity
- Less enjoyable

Donald Trump's most distinctive word in the debate was "they"

US presidential debate, Sep 10th 2024
Selected words, ranked by weighted log-odds ratio



Source: Mark Liberman

Behavioral consequences of second-person pronouns in written communications between authors and reviewers of scientific papers

Pronoun usage's psychological underpinning and behavioral consequence have fascinated researchers, with much research attention paid to second-person pronouns like "you," "your," and "yours." While these pronouns' effects are understood in many contexts, their role in bilateral, dynamic conversations (especially those outside of close relationships) remains less explored. This research attempts to bridge this gap by examining 25,679 instances of peer review correspondence with *Nature Communications* using the difference-in-differences method. Here we show that authors addressing reviewers using second-person pronouns receive fewer questions, shorter responses, and more positive feedback. Further analyses suggest that this shift in the review process occurs because "you" (vs. non-"you") usage creates a more personal and engaging conversation. Employing the peer review process of scientific papers as a backdrop, this research reveals the behavioral and psychological effects that second-person pronouns have in interactive written communications.

Results

Answer your research question

The duration of exposure to running water had a pronounced effect on cumulative seed germination percentages (Fig. 2). Seeds exposed to the 2-day treatment had the highest cumulative germination (84%), 1.25 times that of the 12-h or 5-day groups and 4 times that of controls.

trend

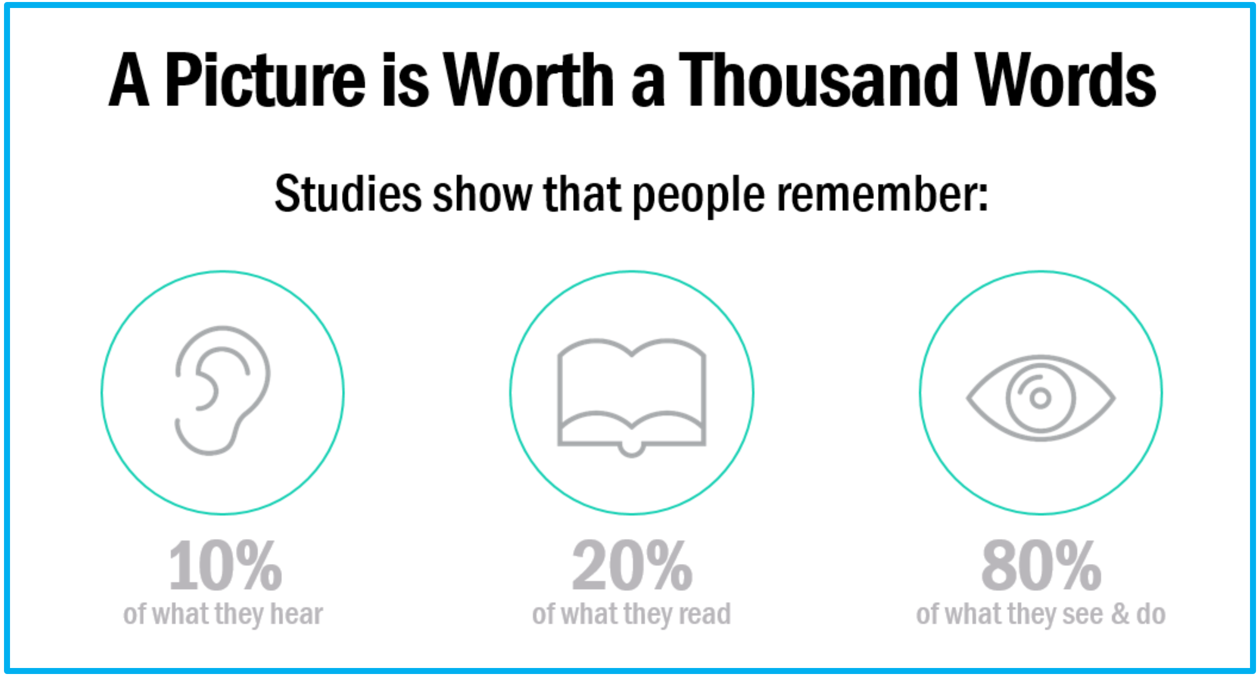
difference

The results of the germination experiment (Fig. 2) suggest that the optimal time for running-water treatment is 2 days. This group showed the highest cumulative germination (84%), with longer (5 d) or shorter (12 h) exposures producing smaller gains in germination when compared to the control group.

optimality

a conceptual model

Results


A Picture is Worth a Thousand Words
Studies show that people remember:
10%
of what they hear
20%
of what they read
80%
of what they see & do

Results

Tables and Figures

- Tables and Figures are assigned numbers separately and in the sequence that you will refer to them from the text.
  - The first Table you refer to is Table 1, the next Table 2 and so forth.
  - Similarly, the first Figure is Figure 1, the next Figure 2, etc.

- When referring to a Figure from the text, "Figure" is abbreviated as Fig.,e.g., **Fig. 1**. Table is never abbreviated, e.g., **Table 1**.

Tables and Figures

- Each Table or Figure must include a brief description of the results being presented and other necessary information in a title.
- No standard rules for writing titles. Grammatically correct.

Figure 1. The main characteristics of the shock absorbers.

Results

Title: brief, sufficiently detailed (journal specific)

- what results are being shown in the graph(s) including the summary statistics plotted
- the organism studied in the experiment (if applicable),
- context for the results: the treatment applied or the relationship displayed, etc.
- location (ONLY if a field experiment),
- specific explanatory information needed to interpret the results shown
- culture parameters or conditions if applicable
- sample sizes and statistical test summaries as they apply.

To an expert reader, the information conveyed in a figure should be clear without needing to consult the main text.

(Nature)

Results

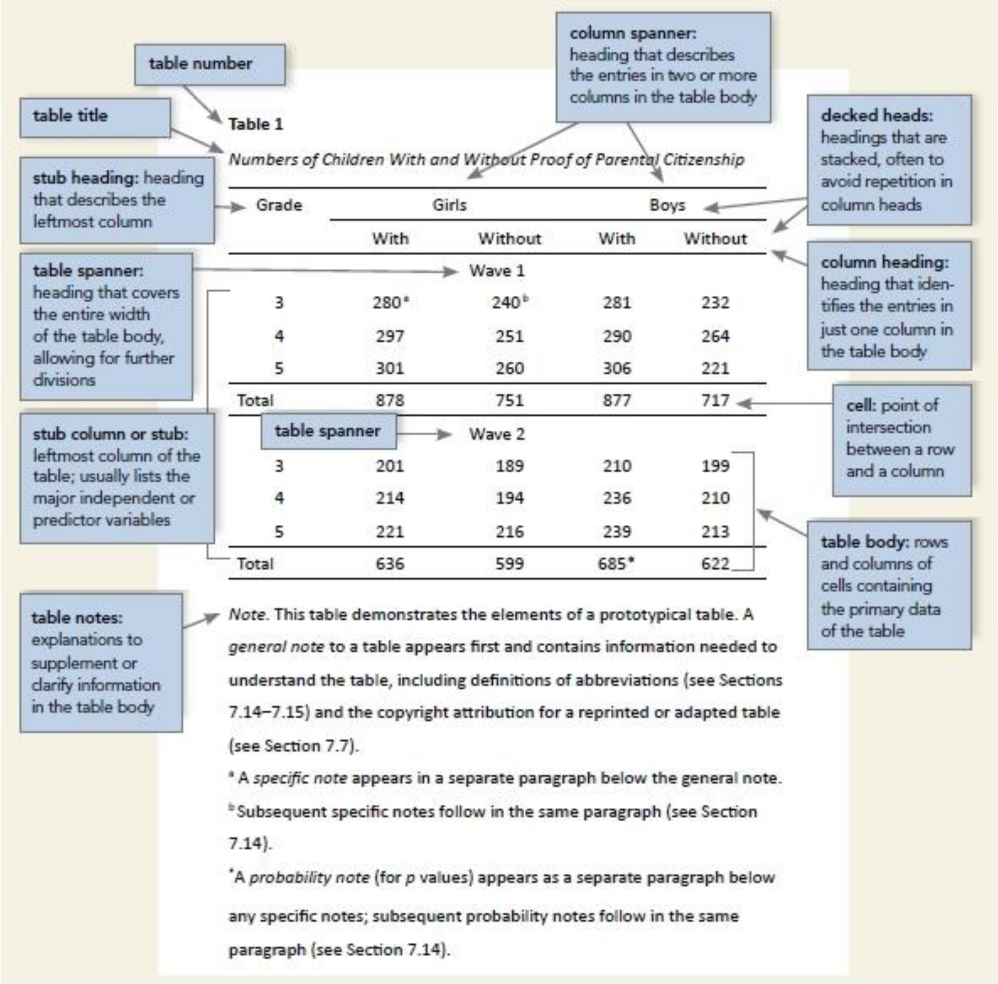


Table 1

Numbers of Children With and Without Proof of Parental Citizenship

Grade	Girls		Boys	
	With	Without	With	Without
		Wave 1		
3	280[a]	240[b]	281	232
4	297	251	290	264
5	301	260	306	221
Total	878	751	877	717
		Wave 2		
3	201	189	210	199
4	214	194	236	210
5	221	216	239	213
Total	636	599	685*	622

Note. This table demonstrates the elements of a prototypical table. A *general note* to a table appears first and contains information needed to understand the table, including definitions of abbreviations (see Sections 7.14–7.15) and the copyright attribution for a reprinted or adapted table (see Section 7.7).

[a] A *specific note* appears in a separate paragraph below the general note.

[b] Subsequent specific notes follow in the same paragraph (see Section 7.14).

*A *probability note* (for *p* values) appears as a separate paragraph below any specific notes; subsequent probability notes follow in the same paragraph (see Section 7.14).

APA

Results

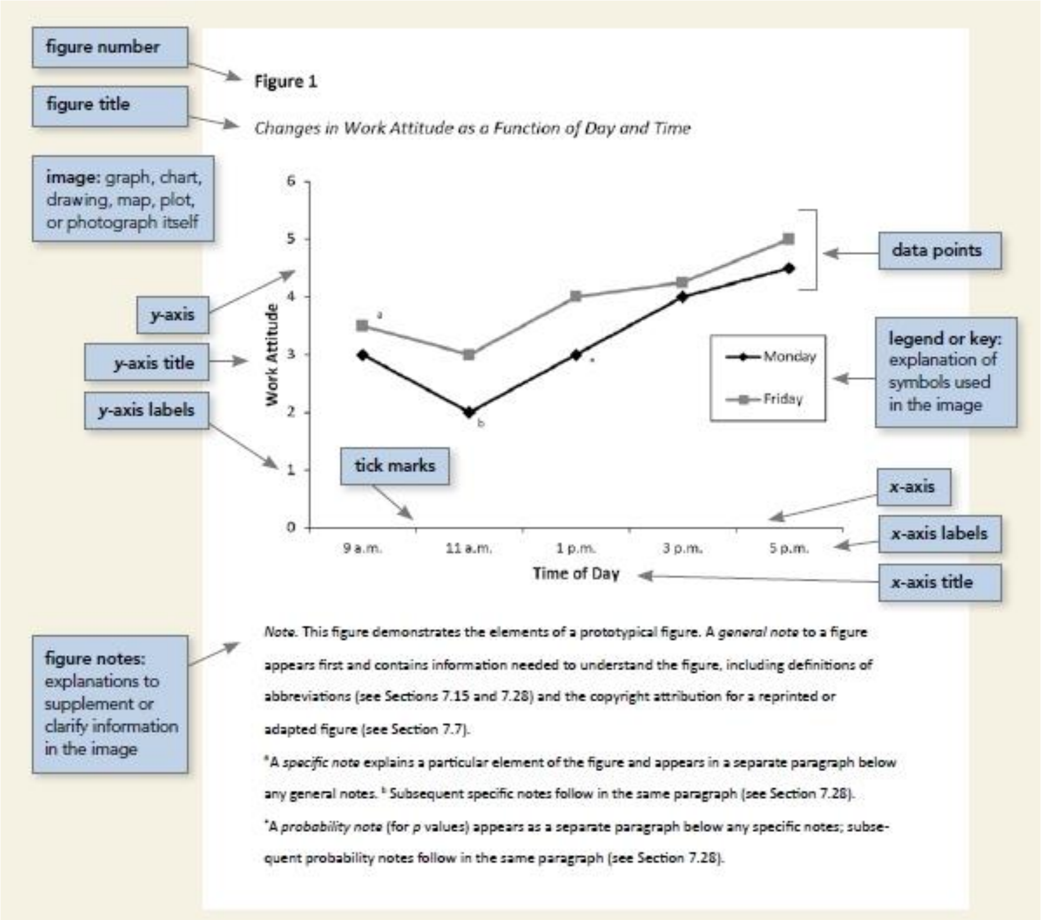


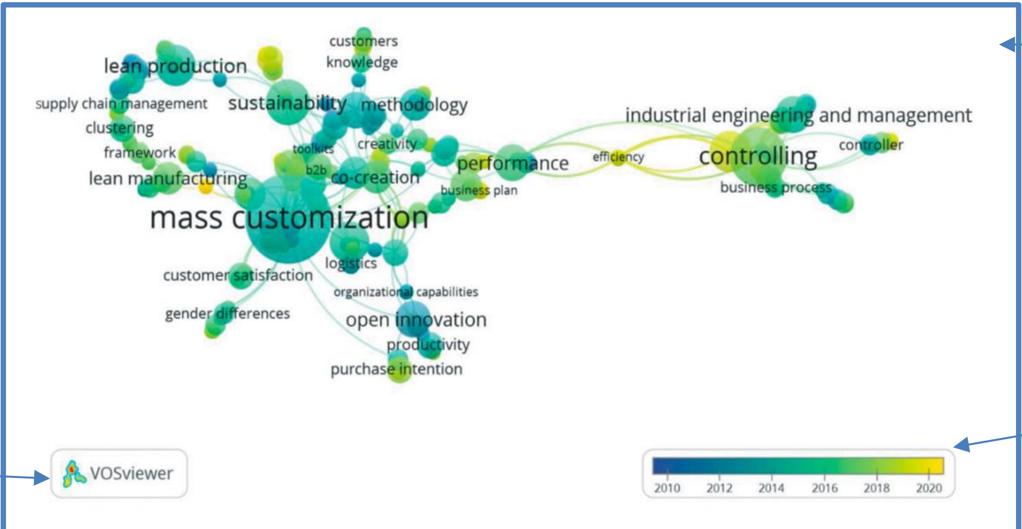
figure number
figure title
image: graph, chart, drawing, map, plot, or photograph itself
y-axis
y-axis title
y-axis labels
figure notes: explanations to supplement or clarify information in the image
data points
legend or key: explanation of symbols used in the image
tick marks
x-axis
x-axis labels
x-axis title
Figure 1
Changes in Work Attitude as a Function of Day and Time
Work Attitude
6
5
4
3
2
1
0
Monday
Friday
9 a.m.
11 a.m.
1 p.m.
3 p.m.
5 p.m.
Time of Day
Note. This figure demonstrates the elements of a prototypical figure. A general note to a figure appears first and contains information needed to understand the figure, including definitions of abbreviations (see Sections 7.15 and 7.28) and the copyright attribution for a reprinted or adapted figure (see Section 7.7).
[a]A specific note explains a particular element of the figure and appears in a separate paragraph below any general notes. [b] Subsequent specific notes follow in the same paragraph (see Section 7.28).
*A probability note (for p values) appears as a separate paragraph below any specific notes; subsequent probability notes follow in the same paragraph (see Section 7.28).

APA

Results

Lead By Example

Image



Legend

Instrument

Figure 2. The most common n-grams by far, colored by the average publication year of papers containing them

Figure title

Figure number

- show an overall trend or “picture” in data;
- convey messages through “shape” rather than the actual numbers;
- allow visual comparisons between data.

Results

A visual bibliometric network shown in Figure 2 was generated by VOSviewer software. It was constructed to graphically display the most common ngrams from all IJIEM papers that are covered by this research. N-gram names are associated with circles, while the size of the circle depends on the frequency of occurrence of that n-gram in the papers. Furthermore, circle color depends on the average year of publication of the papers containing a certain ngram.

Since this research covers papers published in the period between 2010 and 2020, a legend in the bottom right corner of the figure contains a color scale where each shade is used to represent the different years in this period. Brighter colors represent that on average newer papers contain the observed ngram while darker colors mean older papers. Shades of light green and yellow color in Figure 2 indicate that topics like controlling, increasing efficiency and performance, as well as the development of business plans are becoming immensely popular in the years starting from 2018. Similarly, focus on customers and their satisfaction is noticeable in the same period given that n-grams customers, purchase intention, customer satisfaction, and b2b are painted in lighter shades.

Refer to the figure

Instrument

Purpose

Parameters

Explanation of message

Results

4 patterns for referring to non-verbal data

- *Pattern A:*
  - The high rates are shown in Table 1.
- *Pattern B:*
  - The rates were high (see Table 1).
- *Pattern C:*
  - Table 1 shows the high rates.
- *Pattern D:*
  - As shown in Table 1, the rates were high.
  - The rates were high, as shown in table 1.

Discussion

Function

- The function of the Discussion is to interpret your results in light of what was already known about the subject of the investigation, and to explain our new understanding of the problem after taking your results into consideration.
- The Discussion will always connect to the Introduction by way of the question(s) or hypotheses you posed and the literature you cited, but it does not simply repeat or rearrange the Introduction. Instead, it tells how your study has moved us forward from the place you left us at the end of the Introduction.

Discussion

Template

- What are my most important findings？
- Do these findings support what I set out to demonstrate at the beginning of the paper?
- How do my findings compare with what others have found? How consistent are they?
- What is my personal interpretation of my findings?
- What are the limitations of my study? What other factors could have influenced my findings? Have I reported everything that could make my findings invalid?
- Do my interpretations contribute some new understanding of the problem that I have investigated? In which case do they suggest a shortcoming in, or an advance on, the work of others?
- How could my findings be generalized to other areas?
- What possible implications or applications do my findings have?
- What further research would be needed to explain the issues raised by my findings?

Discussion

Organization

One possible way:

- **The beginning part:** The first sentence of the first paragraph should state the importance and the new findings of your research. The first paragraph may also include answers to your research questions mentioned in your introduction section.
- **The middle part:** The middle should contain the interpretations of the results to defend your answers, the strength of the study, the limitations of the study, and an update literature review that validates your findings.
- **The end part:** The end concludes the study and the significance of your research.

(Ghasemi, 2019)

Discussion

Another possible way:

- Discussion of important findings
- Comparison of your results with other published works
- Strength and limitations of the study
- Conclusion and possible implications of your study (including the significance of your study)
- Future research questions based on your findings

(Viera *et al*., 2018)

Discussion

The results and the findings from the conducted research are discussed in this section. The overall impression acquired by analyzing results presented in previous tables and figures is that the IJIEM is an emerging journal with a high potential for growth. The IJIEM plays an active role in the fields of industrial engineering, manufacturing engineering, and production management. Authors of the papers have shifted focus from generalized topics to more specific ones. (p239)

…

Since the most influential IJIEM paper [30] by far deals with the Fourth Industrial Revolution while, at the same time, presented tables show that IT topic, in general, was not very popular in the previous years. A great opportunity for further research could be to explore the possibilities of using artificial intelligence, advanced operational and informational technologies in the execution of managerial and industrial processes [1]. The benefits of exploring this could be huge, given that Kiel et al. [30] claim that it positively affects different aspects of life at the same time – economical, ecological, and social. They state that from the economical aspect main benefits are shorter times needed to finish a job, enhanced productivity and flexibility, increased efficiency, and decreased costs. (p240)

Lead By Example

Summary of the findings: answers to research questions

- Overview
- Current trends
- Further research
- Consistency with other studies

Discussion

Style

- Use the active voice whenever possible in this section.
- Watch out for wordy phrases; be concise and make your points clearly.
- Use of the first person is okay, but too much use of the first person may actually distract the reader from the main points.

Discussion

Approach

- Discuss each in the same sequence as presented in the Results, providing your interpretation of what they mean in the larger context of the problem.

Discussion

Approach

- Use "bridge sentences" that relate the result to the interpretation:

  "The slow response of the lead-exposed neurons relative to controls suggests that...[interpretation]”.
- You must relate your work to the findings of previous studies.
- Use subheadings, if need be, to help organize your presentation.

Structured Discussion

Please ensure that the discussion section of your article comprises no more than five paragraphs and follows this overall structure, although you do not need to signpost these elements with subheadings:

- Statement of principal findings
- Strengths and weaknesses of the study
- Strengths and weaknesses in relation to other studies, discussing important differences in results
- Meaning of the study: possible explanations and implications for clinicians and policymakers
- Unanswered questions and future research

Conclusion?

nature physics

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nature › nature physics › editorials › article

Published: September 2007

Elements of style

Nature Physics **3**, 581(2007) | Cite this article

27k Accesses | **3** Citations | **47** Altmetric | Metrics

We regularly get queries about the minutiae of *Nature Physics* format, but what we really care about is that the papers are clear and accessibly written.

Conclusions are not mandatory, and those that merely summarize the preceding results and discussion are unnecessary (and, for publication in *Nature Physics*, will be edited out).

Conclusion

summary and looking forward

- a very brief revisit of the most important findings pointing out how these advance your field from the present state of knowledge
- a final judgment on the importance and significance those findings in terms of their implications and impact
- possible applications to other areas
- an indication of the limitations of your study
- suggestions for improvements
- recommendations for future

Limitation

- The first limitation of this study is that a time period of only eleven years was taken into account.
- In addition to that, a big part of data preparation for analysis was done manually.
- Also, a potential limitation of this paper could be that the analysis relies solely on data available in the Scopus database.
- Lastly, the first three periods that were selected in the analysis contained time intervals of three years while the last period contained only two years.

- small or limited sample size
- problems encountered when gathering or analyzing data
- confounding variables that were unable to control

Limitation

- The generalizability of the results is limited by…
- The reliability of these data is impacted by…
- Due to the lack of data on x, the results cannot confirm…
- The methodological choices were constrained by…
- It is beyond the scope this study to…

After noting the limitations, reiterate why the results are nonetheless valid for the purpose of answering your research question.

Recommendations

Give concrete ideas for how future work can build on areas that this research was unable to address.

- Further research is needed to establish…
- Future studies should take into account…
- Avenues for future research include…

Reading research papers

Put the sentences in correct order.

1. AI systems are used in a broad range of applications, as varied as connecting people, providing entertainment, and helping create and distribute vaccines.
2. Systems incorporating artificial intelligence (AI) have become an inseparable part of many people's lives, in ways that have positive and negative consequences for users, other people, and society.
3. With the aim of exploring in what sense and to what extent HCAI is human-centered, in this paper we seek to compare HCAI in principle (i.e., as set out in existing guidelines) and in practice (i.e., as evidenced by self-reported developer priorities) with the experiences of AI users.
4. Moreover, it is unclear to what extent these guidelines—and the practices that should follow from attempting to apply them— are truly "human-centered" in the sense of focusing on how AI affects people.
5. However, HCAI guidelines tend to focus on broad concepts such as human values, ethics, and privacy, which can be overly abstract and thus difficult to implement in practice.
6. Reflecting the increasing importance and integration of AI in people's lives, there is a move towards human-centered AI (HCAI), which has the putative goal of placing the user rather than technology at the center of AI development.
7. However, embedding AI in systems can have profound negative consequences —as seen when people are incorrectly denied unemployment benefits on the basis of facial recognition software.

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Introduction

Contextualizing AI’s Impact:
dual-edged nature, encompassing both positive and negative consequences
AI ➡️ HCAI
1.1 Introducing HCAI as a Response

- placing humans at the core of AI systems
- overly abstract, too vague to be helpful

1.2 AI user experiences

- limited focus on user experiences
- less helpful for all users

1.3 Research Questions

- gap： between HCAI principles and their practical implementation
- RQs:
  - Alignment of Priorities: Do developer priorities align with user experiences?
  - Alignment of Principles: Do developer priorities align with HCAI principles?
  - Overall Alignment: Does HCAI, encompassing both principles and practices, align with user experiences?

Methods

2.1. Participants

- sampling method: convenience sampling
- participant groups: seven participant groups, encompassing AI users and developers, a variety of backgrounds
- sample size: 205, with 135 AI users and 70 developers

2.2 Survey items

- open-ended questions
  - users: positive and negative experiences with AI
  - developers: factors consider when designing AI systems

2.3 Coding:

- thematic content analysis
- coding process
- coding scheme: 39

2.4. Comparative Analysis

Results

3.1 Developer Perspectives and User Experiences

3.1.1 Functionality

- key concern for both users and developers
- crucial for both positive and negative experiences, with a greater impact on avoiding negative experiences

3.1.2 Social Impact

- developers prioritize social impact less than users

3.1.3 Understandability vs. Understanding Users

- HCAI: emphasizes the importance of AI being explainable
- users: AI understanding them

3.1.4 Ethics, Privacy, and Security

- developers: prioritize ethics, privacy, and security in AI design
- users: rarely mentioned

3.1.5 Unsure

- limited understanding or exposure to AI systems

Discussion

Key Findings and their Implications for HCAI.

Key findings	Implications for HCAI
➢ Functionality was a defining feature of user experiences with AI and was a key priority for developers	• Making AI systems work *for the user* is a vital part of making them human-centered
➢ Social impact was a defining feature of user experiences with AI, particularly positive experiences	• HCAI requires a greater focus on, and clearer conceptualization of, the impact of AI on users and other people.
➢ Understandability was less important for users than *being understood* by AI	• While understandability, explainability, and interpretability are important for HCAI, it is also important that AI systems make the user feel understood
➢ Ethics, privacy, and security were important developer priorities, but were not key aspects of AI user experiences	• AI policies tend to reflect a focus on the avoidance of harm, which is essential but on its own not sufficient to fulfill the goals of HCAI
➢ Many users could not identify an example of a positive or negative experience with AI	• More research is required to understand the extent to which people are aware of AI systems in their lives

Discussion

- 4.1 Incorporating Human Needs Into HCAI
  - strengthen users’ connections to groups and society
  - maximize the autonomy of users
  - achieve shared goals
  - feel competent
- 4.2 Strengths, Limitations, and Future Research

Conclusion

- key findings
- contribution

Activity

Work on the collocations.

① 一致

② 更关注

③ 标志性特点

④ 与用户相比，开发者在设计 AI 时认为社会影响是一个较低优先级的事项。

① be aligned with

② be more concerned with

③ a defining feature

④ Relative to users, developers perceived social impact to be less of a priority when it comes to designing AI.

Learning achievements

- ✓ Style and function of Results and Discussion
- ✓ Tables+figures
- ✓ Building coherence
- ✓ Collocation